Dorey's Resource Book

Topic 3 – Right-angled triangles

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1 Pythagoras' theorem

ANCHOR QUESTION

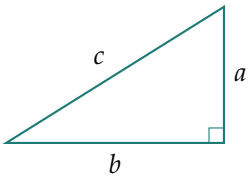
A right-angled triangle has shorter sides 9 and 12. Find the length of the hypotenuse.

SYLLABUS MST-12-S1-04

- Apply Pythagoras' theorem to solve practical problems in two dimensions

NOTES

- The **hypotenuse** is the longest side of a right-angled triangle. It is opposite the right angle.
- **Pythagoras' theorem:** $c^2 = a^2 + b^2$, where c is the hypotenuse.
- To find the hypotenuse, add the squares: $c = \sqrt{a^2 + b^2}$.
- To find a shorter side, subtract the squares: $a = \sqrt{c^2 - b^2}$.
- Write all sides in the same units first. Round only at the end.



WORKED EXAMPLE

A right-angled triangle has shorter sides 8 and 15. Find the length of the hypotenuse, x.

$x^2 = 8^2 + 15^2$ hypotenuse: add the squares

$x^2 = 64 + 225 = 289$

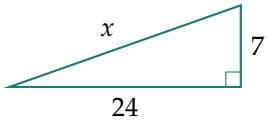
$x = \sqrt{289} = 17$ square root

Tier 1 · Finding the hypotenuse

1. The shorter sides of a right-angled triangle are 3 and 4. Find the hypotenuse.

2. The shorter sides of a right-angled triangle are 5 and 12. Find the hypotenuse.

3. Find x.



Tier 2 · Rounding

4. The shorter sides of a right-angled triangle are 4 and 9. Find the hypotenuse, correct to two decimal places.

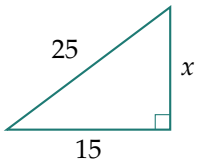
5. The shorter sides of a right-angled triangle are 2.5 and 6. Find the hypotenuse.

6. The shorter sides of a right-angled triangle are 11 and 14. Find the hypotenuse, correct to one decimal place.

Tier 3 · Finding a shorter side

7. A right-angled triangle has a hypotenuse of 10 and one shorter side of 8. Find the other shorter side.

8. Find x.



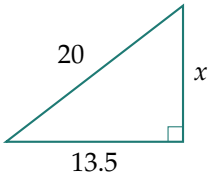
9. A right-angled triangle has a hypotenuse of 17 and one shorter side of 8. Find the other shorter side.

Tier 4 · Shorter side with rounding

10. A right-angled triangle has a hypotenuse of 12 and one shorter side of 7. Find the other shorter side, correct to two decimal places.

11. A right-angled triangle has a hypotenuse of 8.5 and one shorter side of 4.2. Find the other shorter side, correct to two decimal places.

12. Find x, correct to two decimal places.



Tier 5 · Mixed units

13. The shorter sides of a right-angled triangle are 60 cm and 0.8 m. Find the hypotenuse in metres.

14. A right-angled triangle has a hypotenuse of 2.6 m and a shorter side of 100 cm. Find the other side in metres.

15. The shorter sides of a right-angled triangle are 1.5 km and 800 m. Find the hypotenuse in kilometres.

Tier 6 · Two triangles

16. In triangle ABC, D is a point on BC and AD is perpendicular to BC. $AB = 13$, $BD = 5$ and $DC = 16$. Find AD, then find AC.

17. In triangle ABC, AD is perpendicular to BC. $AB = 10$, $BD = 6$ and $DC = 6$. Find AC.

18. A square has sides of 7 cm. Find the length of its diagonal, correct to two decimal places.

Conceptual

19. Use Pythagoras' theorem to explain why the hypotenuse is always the longest side.

20. A triangle has a hypotenuse of 10 and a shorter side of 6. Mia finds the other side as $\sqrt{100 + 36} = 11.66$. Explain her mistake and find the correct answer.

21. A square has a diagonal of 10 cm. Find the side length, correct to two decimal places.

22. Explain why the sides must be in the same units before using Pythagoras' theorem.

23. A composite shape is made of a right-angled triangle with shorter sides 9 cm and 12 cm, and a semicircle drawn on the hypotenuse.

(a) Find the diameter of the semicircle. (2 marks)

(b) Find the perimeter of the composite shape, to the nearest centimetre. (2 marks)

Revision

24. A spinner with 5 equal sections numbered 1 to 5 is spun and a coin is tossed.

(a) How many outcomes are there? (1 mark)

(b) Find $P(\text{head and an even number})$. (1 mark)

(c) Find the probability of NOT getting a head and an even number. (1 mark)

(d) Find the expected number of these results in 300 goes. (2 marks)

25. A garden bed is a trapezium with parallel sides of 6 m and 9 m, which are 4 m apart.

(a) Find the area of the garden bed. (2 marks)

(b) Mulch is spread 0.1 m deep. Find the volume of mulch needed. (1 mark)

(c) Mulch costs \$65 per m^3, plus a \$45 delivery fee. Find the total cost. (2 marks)

26. A tap fills a 9000 L tank at a rate of 15 L/min.

(a) How many hours does it take to fill the tank? (2 marks)

(b) Convert 15 L/min to L/h. (1 mark)

(c) Water costs \$3.20 per kilolitre. Find the cost of the water to fill the tank. (2 marks)

2 Pythagoras in practical problems

ANCHOR QUESTION

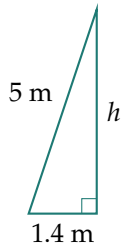
A 5 m ladder leans against a wall with its foot 1.4 *m from the wall. How high up the wall does it reach?*

SYLLABUS MST-12-S1-04

- Apply Pythagoras' theorem to solve practical problems in two dimensions
- Solve practical problems involving Pythagoras' theorem, the trigonometry of right-angled triangles, angles of elevation and depression, and true bearings and compass bearings

NOTES

- Draw a right-angled triangle for the situation and label the lengths you know.
- The hypotenuse is the side opposite the right angle: the ladder, the diagonal, the kite string, the sloping ramp.
- Walking due north then due east (or any two perpendicular directions) makes a right angle.
- Answer the question asked: a saving or a difference may need one more step after Pythagoras.
- State units and round as asked.



WORKED EXAMPLE

A 6 m ladder leans against a wall. Its foot is 2 m from the wall. How high up the wall does it reach, correct to two decimal places?

The ladder is the hypotenuse. — opposite the right angle
$h^2 = 6^2 - 2^2 = 32$ — shorter side: subtract
$h = \sqrt{32} = 5.656\ldots \approx 5.66$ m — round at the end

Tier 1 · Ladders and walls

27. A 3.5 m ladder leans against a wall with its foot 1 m from the wall. How high up the wall does it reach, correct to two decimal places?

28. A ladder reaches 4 m up a wall. Its foot is 1.2 m from the wall. How long is the ladder, correct to two decimal places?

29. A 10 m ladder reaches 8 m up a wall. How far is its foot from the wall?

Tier 2 · Diagonals

30. A rectangular phone screen is 15 cm by 7 cm. Find its diagonal, correct to one decimal place.

31. A gate is 1.8 m wide and 1.2 m high. A diagonal brace is fitted across it. Find the length of the brace, correct to two decimal places.

32. A TV is sold as having a 65-inch diagonal. Its width is 57 inches. Find its height, correct to the nearest inch.

Tier 3 · Directions

33. Ella walks 600 m due south, then 800 m due west. How far is she from her start, in a straight line?

34. A boat sails 12 km due east, then 5 km due north. How far is it from its starting point?

35. A plane flies 230 km due north, then 95 km due east. How far is it from its starting point, correct to the nearest kilometre?

Tier 4 · Comparing distances

36. A rectangular field is 120 m by 50 m. How much shorter is the walk along the diagonal than along two sides?

37. A square courtyard has sides of 20 m. How much shorter is it to cut diagonally across than to walk around two sides, correct to one decimal place?

38. A ramp rises 0.5 m over a horizontal distance of 6 m. The ramp surface is to be covered with non-slip strip costing \$18 per metre. Find the cost, to the nearest dollar.

Conceptual

39. A ladder is moved further from the wall. Explain what happens to the height it reaches.

40. Sam walks 3 km north, then 4 km north-east. He says he is 5 km from his start. Explain why Pythagoras' theorem does not apply.

41. A 4 m ladder must reach 3.9 m up a wall. How far from the wall can its foot be, correct to two decimal places?

42. Explain why the diagonal of a rectangle is always shorter than the sum of its length and width.

43. A plan of a triangular paddock ABC has a scale of $1 : 400$. The angle at B is a right angle.

(a) On the plan, AB measures 15 cm. Find the actual length of AB in metres. (2 marks)

(b) BC is the same length as AB. Find AC, correct to 3 significant figures. (3 marks)

Revision

44. A bag holds 4 green, 6 yellow and 10 blue marbles. One marble is taken at random.

(a) Find $P(\text{yellow})$ in simplest form. (1 mark)

(b) Find the probability that the marble is not blue, as a percentage. (2 marks)

(c) In 60 draws with replacement, how many greens would you expect? (2 marks)

45. A cylindrical can has a diameter of 8 cm and a height of 12 cm.

(a) Find the volume of the can, correct to the nearest cm^3. (2 marks)

(b) Find its capacity in millilitres. (1 mark)

(c) Find the area of the curved surface, to the nearest cm^2. (2 marks)

46. A builder's plan has a scale of 1 : 50. A wall measures 9.6 cm on the plan.

(a) Find the actual length of the wall in metres. (2 marks)

(b) The wall is 2.4 m high. Find its area. (1 mark)

(c) 1 L of paint covers 12 m^2. How many 4 L tins for two coats? (2 marks)

3 The trigonometric ratios

ANCHOR QUESTION

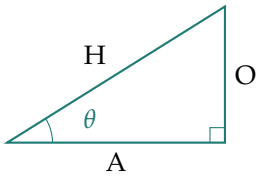
A right-angled triangle has sides 5, 12 and 13. The side opposite θ is 5. Write $\sin\theta$ as a fraction.

SYLLABUS MST-12-S1-04

- Apply trigonometric ratios to find the lengths of unknown sides in right-angled triangles given an angle in both degrees, and degrees and minutes

NOTES

- Name the sides from the marked angle θ. **Hypotenuse (H):** opposite the right angle. **Opposite (O):** across from θ. **Adjacent (A):** next to θ, not the hypotenuse.
- If the marked angle moves to the other acute angle, O and A swap. H never changes.
- $\sin\theta = \dfrac{O}{H}$, $\cos\theta = \dfrac{A}{H}$, $\tan\theta = \dfrac{O}{A}$ (SOH CAH TOA).
- Choose the ratio that uses the two sides in the question: the side you know and the side you want.
- Set the calculator to **degree** mode before finding sin, cos or tan of an angle.



WORKED EXAMPLE

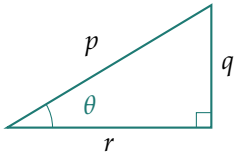
A right-angled triangle has sides 8, 15 and 17. The side opposite θ is 8. Write $\sin\theta$, $\cos\theta$ and $\tan\theta$.

$\text{H} = 17, \text{O} = 8, \text{A} = 15$ — name the sides from θ

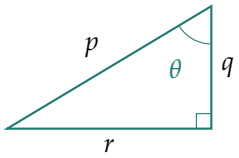
$\sin\theta = \dfrac{8}{17}$, $\cos\theta = \dfrac{15}{17}$, $\tan\theta = \dfrac{8}{15}$ — SOH CAH TOA

Tier 1 · Naming the sides

47. Name the hypotenuse, opposite and adjacent sides for the angle θ.



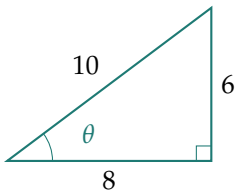
48. Name the hypotenuse, opposite and adjacent sides for the angle θ.



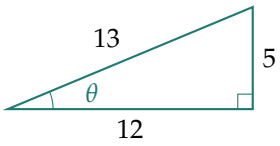
49. In triangle XYZ, the right angle is at Y. For the angle at X, which side is the opposite side?

Tier 2 · Writing one ratio

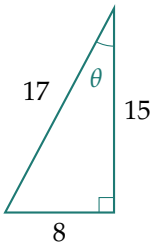
50. Write $\tan\theta$ as a fraction.



51. Write $\cos\theta$ as a fraction.



52. Write $\sin\theta$ as a fraction.



Tier 3 · All three ratios

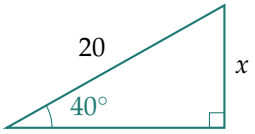
53. A right-angled triangle has sides 3, 4 and 5. The side opposite θ is 3. Write $\sin\theta$, $\cos\theta$ and $\tan\theta$.

54. A right-angled triangle has sides 7, 24 and 25. The side adjacent to θ is 7. Write $\sin\theta$, $\cos\theta$ and $\tan\theta$.

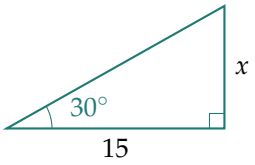
55. A right-angled triangle has sides 20, 21 and 29. The side opposite θ is 20. Write $\sin\theta$, $\cos\theta$ and $\tan\theta$.

Tier 4 · Choosing the ratio

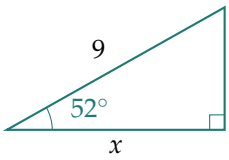
56. Write an equation connecting x and the given side.



57. Write an equation connecting x and the given side.



58. Write an equation connecting x and the given side.



Tier 5 · Using a calculator

59. Find $\sin 50°$, correct to 4 decimal places.

60. Find $\cos 72°$, correct to 4 decimal places.

61. Find $\tan 25°$ and $\tan 65°$, correct to 4 decimal places. Which is greater than 1?

Conceptual

62. Explain why $\sin\theta$ and $\cos\theta$ are always less than 1 for an acute angle in a right-angled triangle.

63. Jake writes $\tan\theta = \frac{A}{O}$. Explain his mistake.

64. In a right-angled triangle, $\sin\theta = \frac{3}{5}$. Find $\cos\theta$ and $\tan\theta$.

65. Explain why the opposite and adjacent sides change when the marked angle moves to the other acute angle.

66. A right-angled triangle has a hypotenuse of 41 and a shorter side of 9, which is opposite the angle θ.

(a) Find the length of the third side. (2 marks)

(b) Write $\tan\theta$ as a fraction. (1 mark)

(c) Which trigonometric ratio of θ is equal to $\frac{40}{41}$? (1 mark)

Revision

67. A ladder 7 m long leans against a wall. Its foot is 1.8 m from the base of the wall.

(a) How high up the wall does the ladder reach, to two decimal places? (2 marks)

(b) Is the height at least 4 times the distance from the wall? Justify. (2 marks)

(c) Write the ratio of distance from the wall to height, in the form $1 : n$. (1 mark)

68. A coin is tossed and a spinner with sections A, B and C is spun.

(a) Draw a tree diagram and list the outcomes. (2 marks)

(b) Find $P(\text{tail and C})$. (1 mark)

(c) Find $P(\text{head or B, or both})$. (2 marks)

69. Priya earns \$1240 a week. Her tax is \$214 a week and the Medicare levy is 2% of her pay.

(a) Find the Medicare levy each week. (1 mark)

(b) Find her net (take-home) pay each week. (2 marks)

(c) Find her net pay for a year (52 weeks). (2 marks)

Solutions

1. $\sqrt{9+16} = \sqrt{25} = 5$

2. $\sqrt{25+144} = \sqrt{169} = 13$

3. $\sqrt{49+576} = \sqrt{625} = 25$

4. $\sqrt{16+81} = \sqrt{97} = 9.848\ldots \approx 9.85$

5. $\sqrt{6.25+36} = \sqrt{42.25} = 6.5$

6. $\sqrt{121+196} = \sqrt{317} = 17.80\ldots \approx 17.8$

7. $\sqrt{100-64} = \sqrt{36} = 6$

8. $\sqrt{625-225} = \sqrt{400} = 20$

9. $\sqrt{289-64} = \sqrt{225} = 15$

10. $\sqrt{144-49} = \sqrt{95} = 9.746\ldots \approx 9.75$

11. $\sqrt{72.25-17.64} = \sqrt{54.61} = 7.389\ldots \approx 7.39$

12. $\sqrt{400-182.25} = \sqrt{217.75} = 14.756\ldots \approx 14.76$

13. 0.6 m and 0.8 m. $\sqrt{0.36+0.64} = 1$ m

14. $\sqrt{6.76-1} = \sqrt{5.76} = 2.4$ m

15. 1.5 km and 0.8 km. $\sqrt{2.25+0.64} = \sqrt{2.89} = 1.7$ km

16. $AD = \sqrt{169-25} = 12$. $AC = \sqrt{144+256} = \sqrt{400} = 20$

17. $AD = \sqrt{100-36} = 8$. $AC = \sqrt{64+36} = 10$

18. $\sqrt{49+49} = \sqrt{98} = 9.899\ldots \approx 9.90$ cm

19. $c^2 = a^2 + b^2$, so c^2 is bigger than a^2 and bigger than b^2. That means c is longer than both a and b.

20. She added instead of subtracting. A shorter side cannot be longer than the hypotenuse. $\sqrt{100-36} = 8$.

21. $s^2 + s^2 = 100$, so $s^2 = 50$ and $s = \sqrt{50} = 7.071\ldots \approx 7.07$ cm.

22. The theorem adds and subtracts the squares of the lengths, which only makes sense when they are measured in the same unit.

23. (a) $\sqrt{81+144} = \sqrt{225} = 15$ cm. (b) Arc $= \frac{1}{2} \times \pi \times 15 = 23.56\ldots$ cm. Perimeter $= 9 + 12 + 23.56 = 44.56 \approx 45$ cm.

24. (a) $5 \times 2 = 10$. (b) H2, H4: $\frac{2}{10} = \frac{1}{5}$. (c) $1 - \frac{1}{5} = \frac{4}{5}$. (d) $300 \times \frac{1}{5} = 60$.

25. (a) $\frac{4}{2} \times (6+9) = 30$ m^2. (b) $30 \times 0.1 = 3$ m^3. (c) $3 \times \$65 + \$45 = \$240$.

26. (a) $9000 \div 15 = 600$ min $= 10$ h. (b) $15 \times 60 = 900$ L/h. (c) 9000 L $= 9$ kL. $9 \times \$3.20 = \$28.80$.

27. $\sqrt{12.25 - 1} = \sqrt{11.25} = 3.354\ldots \approx 3.35$ m

28. $\sqrt{16 + 1.44} = \sqrt{17.44} = 4.176\ldots \approx 4.18$ m

29. $\sqrt{100 - 64} = 6$ m

30. $\sqrt{225 + 49} = \sqrt{274} = 16.55\ldots \approx 16.6$ cm

31. $\sqrt{3.24 + 1.44} = \sqrt{4.68} = 2.163\ldots \approx 2.16$ m

32. $\sqrt{4225 - 3249} = \sqrt{976} = 31.2\ldots \approx 31$ inches

33. $\sqrt{360\,000 + 640\,000} = \sqrt{1\,000\,000} = 1000$ m

34. $\sqrt{144 + 25} = 13$ km

35. $\sqrt{52\,900 + 9025} = \sqrt{61\,925} = 248.8\ldots \approx 249$ km *cf. 2024 Std 1 HSC Q14(a)*

36. Two sides $= 170$ m. Diagonal $= \sqrt{14\,400 + 2500} = 130$ m. Saving $= 40$ m.

37. Two sides $= 40$ m. Diagonal $= \sqrt{800} = 28.28\ldots$ m. Saving ≈ 11.7 m.

38. Length $= \sqrt{36 + 0.25} = 6.020\ldots$ m. Cost $= 6.020\ldots \times 18 = \$108.37 \approx \$108$.

39. The ladder (hypotenuse) stays the same length. A longer base means $h^2 = c^2 - b^2$ is smaller, so it reaches lower.

40. North and north-east are not at right angles, so the path does not make a right-angled triangle.

41. $\sqrt{16 - 15.21} = \sqrt{0.79} = 0.888\ldots \approx 0.89$ m

42. The diagonal is a straight line between two corners; walking along two sides is a longer path between the same points.

43. (a) $15 \times 400 = 6000$ cm $= 60$ m. (b) $AC = \sqrt{60^2 + 60^2} = \sqrt{7200} = 84.85\ldots \approx 84.9$ m. *cf. 2025 Std 1 HSC Q20*

44. (a) $\frac{6}{20} = \frac{3}{10}$. (b) $1 - \frac{10}{20} = \frac{1}{2} = 50\%$. (c) $60 \times \frac{4}{20} = 12$.

45. (a) $\pi \times 4^2 \times 12 = 603.18\ldots \approx 603$ cm^3. (b) 603 mL. (c) $2 \times \pi \times 4 \times 12 = 301.59\ldots \approx 302$ cm^2.

46. (a) $9.6 \times 50 = 480$ cm $= 4.8$ m. (b) $4.8 \times 2.4 = 11.52$ m^2. (c) Two coats $= 23.04$ m^2. $23.04 \div 12 = 1.92$ L, so one 4 L tin.

47. H $= p$, O $= q$, A $= r$

48. H $= p$, O $= r$, A $= q$

49. YZ (across from X)

50. $\frac{6}{8} = \frac{3}{4}$

51. $\frac{12}{13}$

52. O = 8 (bottom side), H = 17: $\frac{8}{17}$

53. $\frac{3}{5}$, $\frac{4}{5}$, $\frac{3}{4}$

54. O = 24: $\frac{24}{25}$, $\frac{7}{25}$, $\frac{24}{7}$

55. $\frac{20}{29}$, $\frac{21}{29}$, $\frac{20}{21}$

56. O and H: $\sin 40^\circ = \frac{x}{20}$

57. O and A: $\tan 30^\circ = \frac{x}{15}$

58. A and H: $\cos 52^\circ = \frac{x}{9}$

59. 0.7660

60. 0.3090

61. $\tan 25^\circ = 0.4663$, $\tan 65^\circ = 2.1445$. $\tan 65^\circ$ is greater than 1.

62. Both have the hypotenuse on the bottom, and the hypotenuse is the longest side, so the fraction is less than 1.

63. Tan is opposite over adjacent: $\tan\theta = \frac{O}{A}$ (TOA). He has the fraction upside down.

64. O = 3, H = 5, so A = $\sqrt{25-9} = 4$. $\cos\theta = \frac{4}{5}$, $\tan\theta = \frac{3}{4}$.

65. The opposite side is the one across from the marked angle, so a different angle has a different side across from it. The hypotenuse stays the same.

66. (a) $\sqrt{1681-81} = \sqrt{1600} = 40$. (b) $\tan\theta = \frac{9}{40}$. (c) $\cos\theta$.

67. (a) $\sqrt{49-3.24} = \sqrt{45.76} = 6.764\ldots \approx 6.76$ m. (b) $4 \times 1.8 = 7.2$ m. The height is only 6.76 m, so no. (c) $1.8 : 6.76 \approx 1 : 3.76$.

68. (a) HA, HB, HC, TA, TB, TC. (b) $\frac{1}{6}$. (c) HA, HB, HC, TB: $\frac{4}{6} = \frac{2}{3}$.

69. (a) $0.02 \times \$1240 = \$24.80$. (b) $\$1240 - \$214 - \$24.80 = \$1001.20$. (c) $52 \times \$1001.20 = \$52\,062.40$.